

U2-D4 Homework – Negative Exponents and QUIZ Review**Name:**Simplify completely. [The fact that you can't leave negative exponents is *implied*.]

1. $x^{-2} \cdot x^{-3}$

2. $\frac{x^{-5}}{x^{-12}}$

3. $\frac{x^3 y^9}{x^6 y^{-2}}$

4. $\frac{x^{-4} y^5}{x^2 y^8}$

5. $x^3 (x^4)^{-2}$

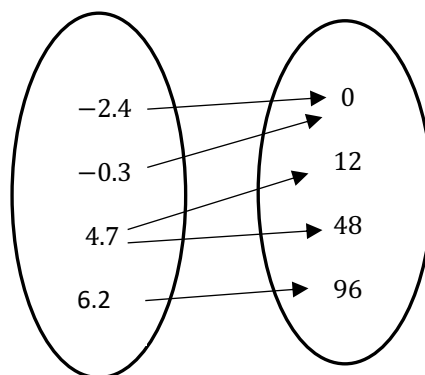
6. $\left(\frac{x^2 y}{x^{-5} y^7} \right)^3$

Determine whether each relation is a function and state domain and range.

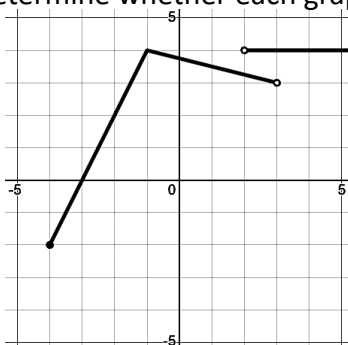
7.

Character	Speed
Mario	5
Luigi	3
Princess	4
Toad	6

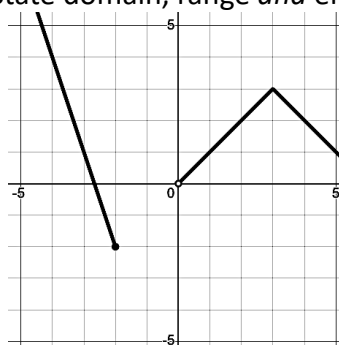
8.

Determine whether each graph represents a function. State domain, range *and* end behavior.

9.



10.



Evaluate with a graphing calculator. Round to nearest 1000th.

11. $\frac{3^4+7}{4}$

12. $\sqrt{-13+71} - 5\sqrt{2}$

Solve on a graphing calculator. Round to nearest 1000th. NOTE: You don't know what "log" means, but it *is* a button on the TI-84. It can solve things you don't even understand!

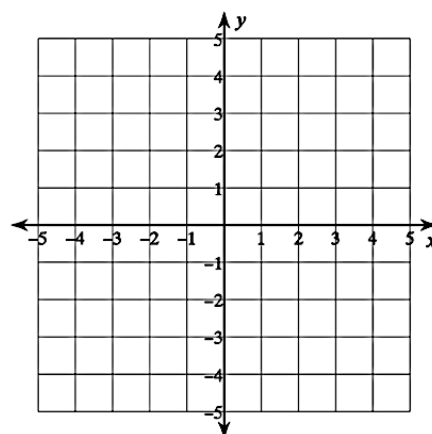
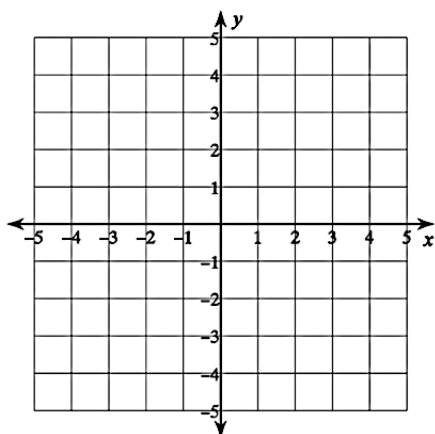
13. $-\frac{2}{3}x + 5 = -4x - 3$

14. $x - 3 = \log(x + 5) + 3$

Graph each absolute value function using at least five points (fewer permitted if there's no space). State the vertex and right slope, as well as the domain, range and end behavior.

1. $y = -|x| + 3$

2. $h(x) = \frac{1}{3}|x - 1| - 2$



3. $y = |x + 3|$

4. $h(x) = -2|x + 2| + 4$

